

Intestinal flora variation reflects the short-term damage of microplastic to the intestinal tract in mice

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ARTICLE INFO

Edited by Dr. Caterina Faggio

Keywords:

Microplastic

Intestinal flora

Biological effect

Short-term exposure

ABSTRACT

The potential toxicity of microplastic (MPs) to organisms has attracted extensive attention. However, due to the subacute toxicity of MPs, the biological effect is hard to verify in short-term exposure experiment. Here, by tracking the dynamics of gut microbes, mice model was utilized to evaluate the toxicity of compositional MPs (PE, PET, PP, PS and PVC). After 7 days digestive exposure, the physiological indicators were normal as the control group that the body weight and serum cholesterol levels were insignificant change. Whereas, through histopathological examination, all the treatment groups suffered colon tissue damage, among which PS had the most inflammatory cells. Moreover, the high-throughput sequencing results revealed great variation of intestinal flora in treated mice. The ratio of Bacteroidetes and Firmicutes in PE, PET and PP treatment groups heighten, and the relative abundance of Ruminococcaceae and Lachnospiraceae increased significantly at family levels. At the genus level, *Alistipes* bacteria in PS treatment group significantly decreased that is associated with obesity risk. It indicated that MPs induced inflammatory response would further interfere the dynamics of intestinal flora causing health effect in living organisms. This work shed light on MPs toxicity in short-term exposure and supplied research paradigm of MPs health risk assessment.

1. Introduction

The plastic pollution has boosted rapidly worldwide, mainly due to the large-scale production and emission in recent decades (Bouwmeester et al., 2015). Some of the waste plastics remaining in the environment become plastic particles through physical friction and chemical aging, forming the microplastic (MPs) of less than 5 mm in diameter (Law and Thompson, 2014). Microplastics are everywhere that can be ingested and accumulated in vivo (Lenaker et al., 2019). Previous research has succeed to detect MPs in various species, such as mussel and fish, and has also been observed in terrestrial organisms of chickens and birds (Browne et al., 2008; Markic et al., 2019; Watts et al., 2014; Zhao et al., 2016). Through the food chain, MPs can be transferred into higher organisms, even in human bodies, which have posed a potential biological toxicity, ecological threat, and human health.

Scientific evidence has shown that animals exposed to microplastic

produce a variety of adverse reactions, mainly from survival rate, oxidative stress and immune response reproductive capacity (Ju et al., 2019; Wan et al., 2019; Wang et al., 2021). In general, biological effects have been validated mainly by indoor biological evaluation models. The juvenile large yellow croaker *Larimichthys crocea* exposed to microplastic could interfere the enzyme activities (LPS, TRS, AMS, LZM) in the intestinal tract (Gu et al., 2020). When zebrafish larvae were treated with microplastic, the metabolism of zebrafish larvae was markedly disturbed in triglyceride (TG), total cholesterol (TC), non-esterified fatty acids (NEFA), total bile acid (TBA), glutamic acid (GLU) and pyruvate (Han et al., 2021). In addition, the offspring metabolism was disrupted in MPs-exposed mice, especial for pregnant mice, with significant changes in 11 and 15 metabolites under 0.5 and 5 μ m MPs treatment groups (Luo et al., 2019). Even so, microplastics are subtoxic pollutants that require prolonged exposure showing the physiological and biochemical effect on animals. Whereas, short-term exposure to

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<https://doi.org/10.1016/j.ecoenv.2022.114194>

Received 26 July 2022; Received in revised form 7 October 2022; Accepted 13 October 2022

Available online 14 October 2022

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microplastic is difficult to show significant changes in physical signs or biological effects, which makes it impossible to accurately grasp the potential toxicity of microplastic. Therefore, more sensitive indicators need to be excavated to assess the early influence of microplastic exposure.

The combination of microorganisms present in organism gut is called the gut microbiome (Sekirov et al., 2010). It is health-involved in the regulation of host physiological functions of digestion, immune regulation and neurodevelopment (Backhed et al., 2004; Kanauchi et al., 2013; J. Xu et al., 2003). More than 3.3 million genomes of gut microbes are existing in organism that are highly dynamic and evolutionary (Qin et al., 2010). They are susceptible to environmental substance and always selected as toxicological target of various pollutants (Jin et al., 2017; Lu et al., 2019; Rosenfeld, 2017). For example, gut microbiome of earthworm have been used to monitor the heavy metal toxicity in arsenic-exposed soil environment (Hong-Tao et al., 2019). Hence, gut microbes are preferable biotic response indicator for toxic evaluation of pollutants in organism.

Faced with subtoxic characteristic of MPs, we designed a sensitive evaluation model to monitor the short-term effect of multicomponent microplastic. The PE, PET, PP, PS, PVC microplastics were used as target toxicants, and mice were selected as toxicity exposure models for 7 days of continuous feeding. Through quantitative analysis of histopathological damage in mice, the dynamic changes of intestinal microbe in mice were observed, and its biological effects were reflected. It may help provide new toxicity assessment for the potential health risks of microplastic in mammals.

2. Materials and methods

2.1. Chemicals, animals and experimental solutions

Polyethylene (PE), polypropylene (PP), polystyrene (PS), polyvinyl chloride (PVC) and polyethylene terephthalate (PET) were purchased from Baoqian Plastics Co., LTD (Dongguan, China). Prior to start experiment, FT-IR (Nicolet-IS10, Thermo Scientific, USA) and SEM (Quanta 450FEG, FEI, USA) were used to identify the functional groups and morphology of microplastics (Fig. S1). The size and shape of MPs were determined by scanning electron microscopy (SEM), and the microplastics used in this study have irregular shapes and rough surfaces. Further analysis showed that the diameters of PE, PET, PP, PS and PVC mainly concentrated in the range of 150 ~ 300 μm . Among the five microplastics, the size of PET microplastics is slightly larger and that of PS microplastics is slightly smaller. By Fourier transform infrared spectroscopy (FT-IR), we obtained the spectrograms of five microplastics in supporting information Fig. S1 and the peaks of the five microplastics components were consistent with their respective spectral characteristics.

Kunming mice aged 8 weeks were selected and purchased from Hunan Slyke Jingda Experimental Animal Co., LTD (Changsha, China). The culture conditions of the animal house were maintained at 12:12 h day and night cycle, and the temperature was room temperature. Water and food were freely consumed throughout the animal experiment. Before begin experiment, all mice were randomly divided into six groups of three. The mice were given PE, PP, PS, PVC and PET (20 mg/mL) by gavage once a day (0.2 mL/day), respectively. The control group was orally given 0.2 mL ddH₂O. The microplastic solution in the treatment

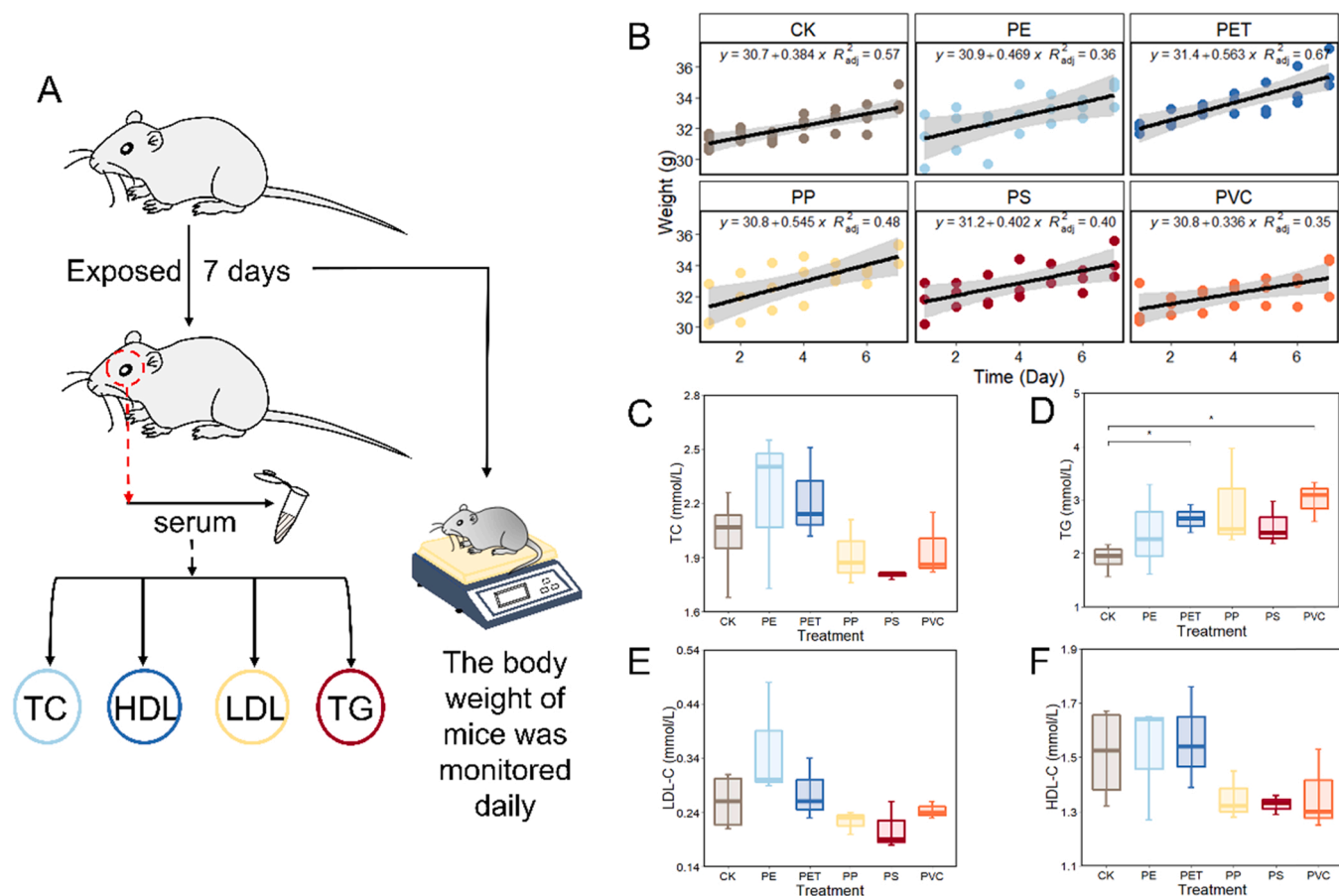


Fig. 1. The schematic diagram of detection of physiological and biochemical indexes of mice (A). The fitting diagram of body weight change in mice (B). Dots represent individual measured values, black lines represent the linear regression, and gray area represents its 95% confidence interval. The TC (C), TG (D), LDL-C (E), HDL-C (F) level of mice after one week of microplastic exposure. The blood samples were collected after sacrifice, and the data were expressed as mean $\pm$ SEMs.

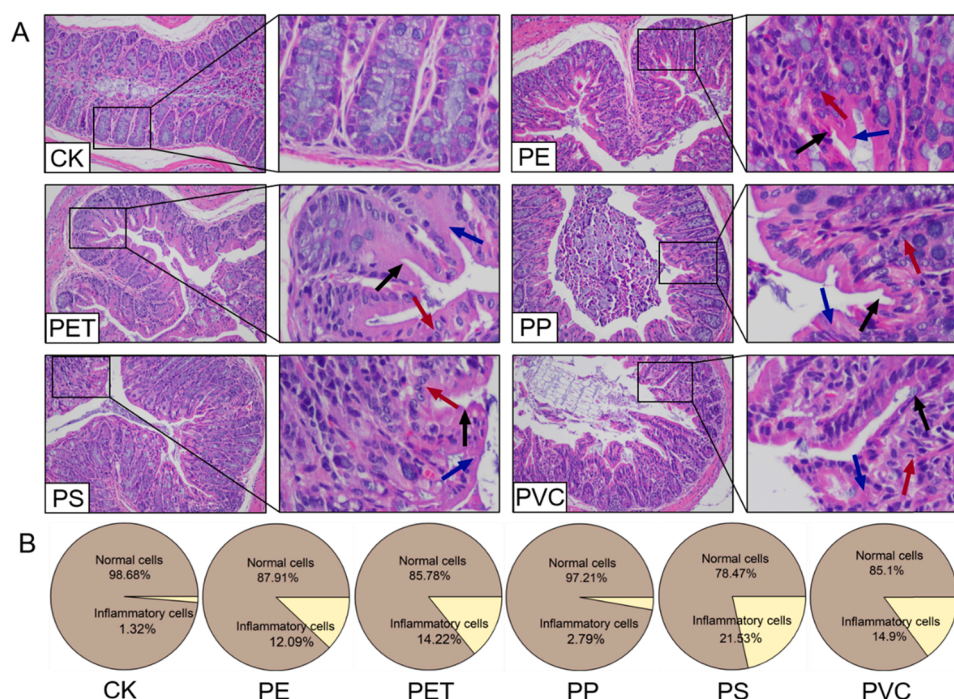


Fig. 2. Effect of microplastic on colon tissue pathological. Representative images of colon tissue with H&E (200 ×) (A). Red arrow: inflammatory cells. Black arrow: crypt surface irregular. Blue arrow: reduced goblet cells. The ratio of normal cells to inflammatory cells in mouse colon tissue in all groups (B).

group was prepared and used every day, and the solution was mixed evenly by ultrasonic shock before gavage. The dosages of the microplastics were determined according to previous studies (Jin et al., 2021b,a; Mu et al., 2022). The exposure lasted for one week and the weight of the mice was measured daily. Stools samples were collected from mice before and during the experiment and frozen at -80°C . All experiments were in accordance with the experimental animal welfare ethics of The Second Xiangya Hospital of Central South University, and were approved to carry out animal research and experiment, and the experimental ethics code is 2020479.

2.2. Histopathological evaluation and analysis of the colon

After experimental period (fasting for 12 h before execution), all mice were euthanized with cervical dislocation method. The part of colon tissues of mice was taken for histopathological examination. The details are shown in the Text S2. Briefly, the histopathological examination followed eight steps: wash, transversely cut, fix, embed, slice, stain, scan and observe.

The remaining colon tissue was weighed and pulverized with a homogenizer under ice bath conditions. The mixture was put into a centrifuge tube and centrifuged at 8000 rpm at 4°C for 10 min. The supernatant was collected to measure physiological indicators, including superoxide dismutase (SOD), glutathione (GSH), malondialdehyde (MDA) and peroxidase (POD). The detailed methods about the determination of SOD, GSH, MDA and POD are shown in Texts S3–S6, respectively. Assay kits were purchased from Beijing Solarbio Science & Technology Co., LTD. (Beijing, China).

2.3. Evaluation of blood lipid biochemical index

Before euthanizing these mice, the serum samples were collected. The biochemical indexes of blood lipid in mice were carried out using automatic biochemical analyzer (FAITH-1000, Nanjing, China), including total cholesterol (TC), triglyceride (TG), low-density lipoprotein cholesterol (LDL-C) and high-density lipoprotein cholesterol (HDL-C).

2.4. DNA extraction, PCR amplification and high-throughput sequencing

According to the manufacturer's instructions, total microbial genomic DNA samples were extracted from the feces of mice using DNA extraction kits (Agilent, China) and stored at -20°C until further analysis. The 16 s rRNA V3-V4 region of bacteria was obtained by using universal primers F515 and R806 to amplify DNA. PCR amplification conditions were as follows: 94°C for 5 min; 30 cycles of 94°C for 30 s, 52°C for 30 s, and 72°C for 30 s; and 72°C for 10 min. Based on the manufacturer's recommendations, the NEBNext® Ultra™ DNA Library Prep Kit from Illumina® (New England Biolabs, MA, USA) was used to generate the sequencing library and index code was added. The quality of the library was evaluated using Qubit® 2.0 fluorescence analyzer (Thermo Fisher Scientific, MA, USA) and Agilent Bioanalyzer 2100 system (Agilent Technologies, Waldbron, Germany). Sequencing was then performed on IlluminaHiseq2500 platform (Guangdong Magigene Biotechnology Co., Ltd.). Details of sequencing analysis were presented in Text S7.

2.5. Statistical analysis

All data are expressed as mean $\pm$ SD. The data were analyzed for statistical significance with SPSS v26.0 (IBM, Chicago, USA). Differences between groups were tested by One-way analysis of variance to determine statistical significance among groups, and $p < 0.05$ was considered statistically significant. Data visualization was performed by R3.6.1 and Origin 2018 (OriginLab, USA). Histopathological quantitative analysis was performed by software QuPath-0.3.2 (Edin, UK).

3. Results

3.1. Effects of PS-MPs exposure on body weight and serum TC, TG, LDL-C and HDL-C in mice

A schematic of the physiological state of a mouse is shown in Fig. 1A. We monitored body weight and serum cholesterol levels to understand the physiological state of mice exposed to microplastics. Fig. 1B presents

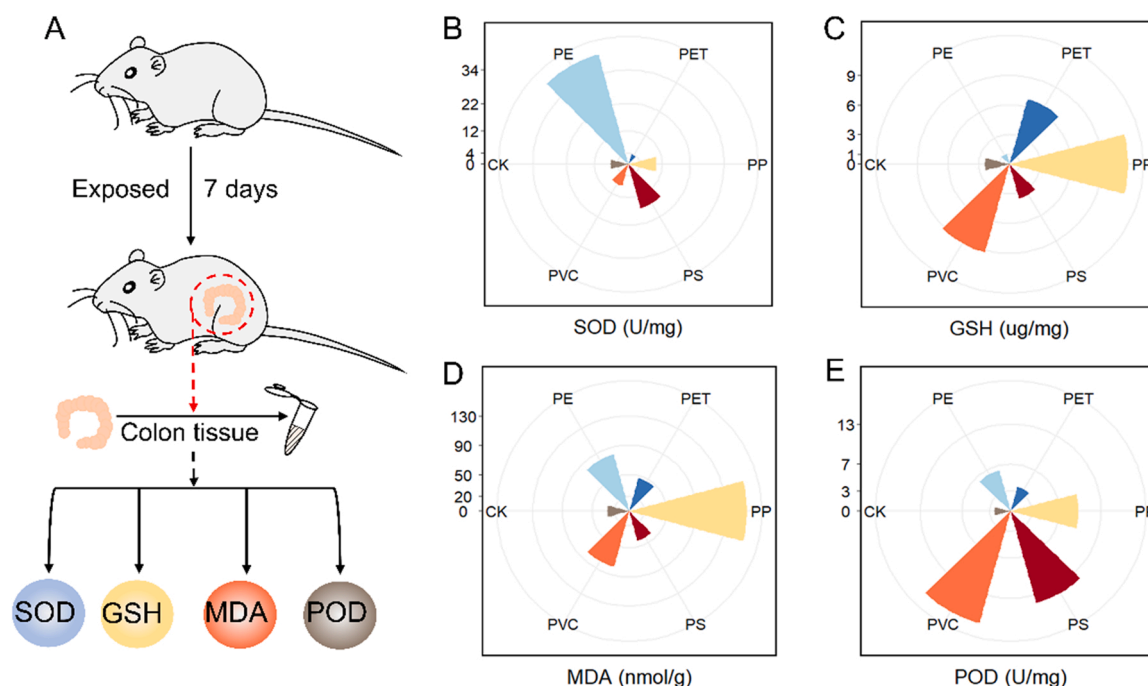


Fig. 3. Schematic diagram of oxidative stress measurement in mice (A). SOD (superoxide dismutase) (B), GSH (glutathione) (C), MDA (malonic dialdehyde) (D), POD (peroxidase) (E).

the changes in body weight over the course of exposure (7 days) in all treatment groups by linear fitting. The slanted lines in each treatment group represent the slope of weight gain, and the higher the slope, the more weight gain. It was observed that among all treatment groups, the maximum slope of PET treatment and PP treatment was 0.563 and 0.545, respectively, which was significantly higher than that of CK (0.384). The results of statistical analysis showed that in all treatment groups, the weight of mice in PP and PET treatment groups increased significantly after 7 days of exposure compared with the first day, and the weight growth rate of mice was also the largest among all treatment groups (Fig. S2). We then analyzed the serum cholesterol levels of the mice. Among the four serum indexes, only TG content showed significant change between the treatment group and the control group through microplastic exposure. As shown in Fig. 1D, TG concentration in PET group and PVC group was significantly higher than that in control group ($P < 0.05$). In the other measures (TC, HDL-C and LDL-C), there were no significant differences between the treatment and control groups (Fig. 1C and E-F).

3.2. The colon histopathology of mice after MPs exposure

To better understand the effects of microplastic on mice, colon tissue was examined using histological stain. H&E staining is widely used in tissue specimens and can quickly evaluate the tissue morphology of samples. Fig. 2A shows the H&E staining of colonic tissue from mice in all treatment groups, magnified 100 times on the left and 200 times on the right under a light microscope in the same treatment group. In this illustration, the colons of control mice are structurally intact, with normal crypts and cell morphology. However, compared with the control group, the colons of mice in all treatment groups showed histopathological damage. In addition, there were signs consistent with the pathology of colitis, including a decline in goblet cells and inflammatory infiltration. And the map of inflammatory cells in colon tissue showed the same damage situation (Fig. S3). To intuitively compare the damage of colon tissues, quantitative analysis of inflammatory cells was conducted (Fig. 2B). We found that the PS treatment group had the most severely histopathological damage among all treatment groups, with

inflammatory cells accounting for 21.53% in all cells. The treatment group of PP had the least inflammatory cells (2.79%), which is slightly higher than the control group (1.47%). The order of inflammatory cells ratio was PS>PVC>PET>PE>PP>CK.

3.3. The colonic oxidative stress in mice after MPs exposure

Next, the oxidative stress in mouse colon tissue was assessed after MPs exposure, including the enzymatic activity of SOD, GSH, MDA and POD (Fig. 3A). SOD concentration in PE treatment group was about 10 times of that in control group (40.99 U/mg), which was much higher than that in other treatment groups (Fig. 3B). However, as shown in Fig. 3C, the concentration of GSH in PE treatment group was only 1.10 U/mg, which was the lowest among all treatment groups. The vary of SOD and GSH contents in mice after PE exposure were completely different from those in the control group. After exposure to five kinds of microplastics, MDA concentration in each treatment group increased significantly compared with the control group (Fig. 3D), and the concentration of PP treatment group heighten the most. Similarly, as shown in Fig. 3E, POD concentration in colon tissues of mice in the five treatment groups were also much higher than that in the control group, and the POD level in the PVC group was the highest among the five treatment groups. Five microplastics were exposed to mice, it was found that the concentrations of SOD, GSH, MDA and POD in the colon of mice were raised in the exposure of PP, PS and PVC microplastics compared with the control group (Fig. 3B-E). In general, all the mice in each MPs exposure suffered oxidative stress response to some degree.

3.4. Effects of MPs on gut microbiota diversity in mice

Shannon dilution curve is an index reflecting the microbial diversity of samples. The microbial diversity index of samples under different sequencing quantities is represented by the microbial diversity index curve under different sequencing depths. As the sequence depth deepen, the Shannon curve of observed species tended to be flat (Fig. S4), indicating that the sequence could fully cover the bacteria in the sample. Alpha diversity indices (Chao1, Shannon and Simpson) were calculated

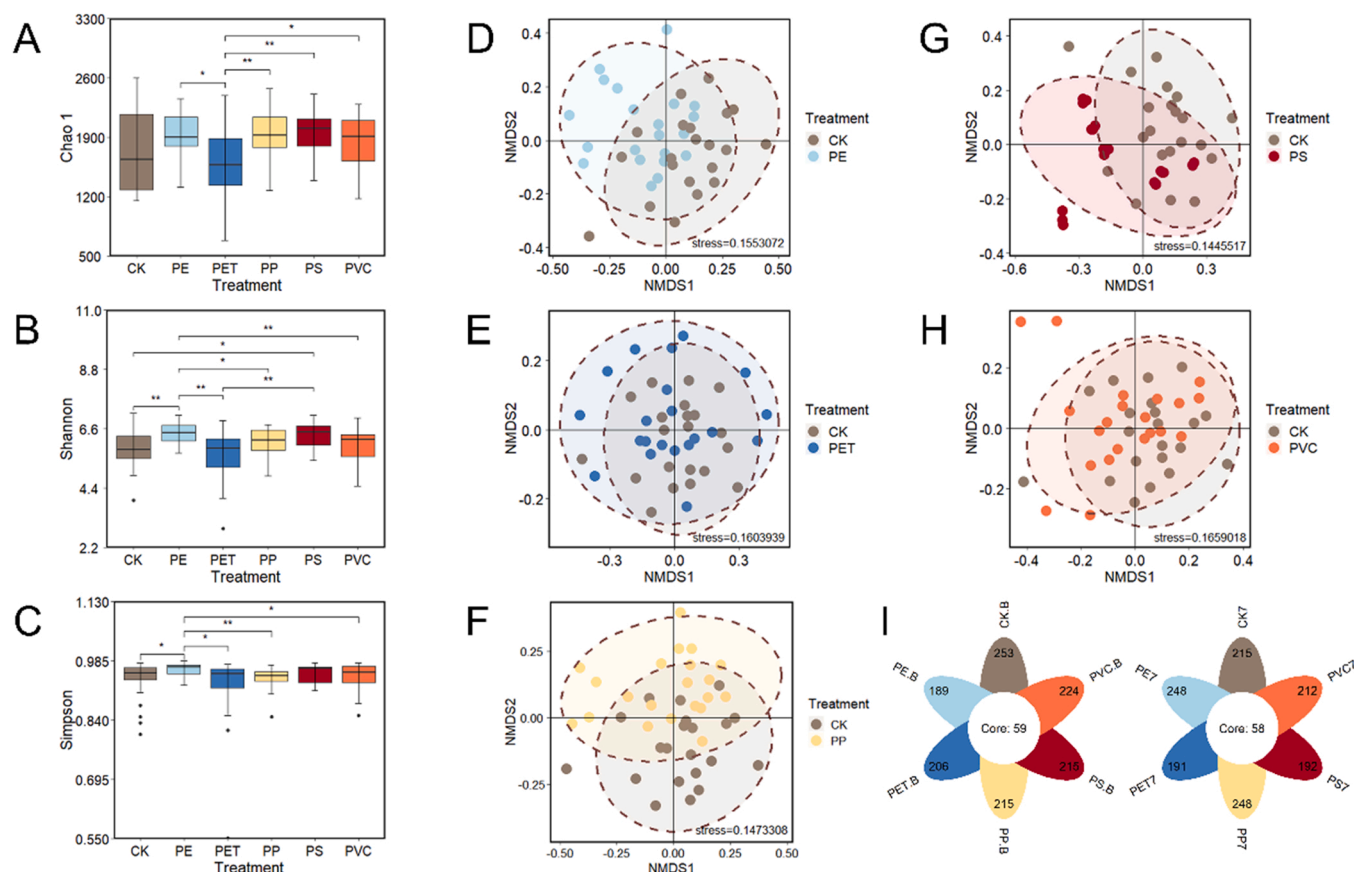


Fig. 4. Intestinal α -diversity of all exposed groups included Chao1 index, Shannon index and Simpson index (A). The intestinal microflora of mice was analyzed by NMDS based on Bray-Curtis distance (B). Venn diagram(C), number of OTUs before exposure (B) and on day 7 of exposure.

for assessing the species diversity of individual samples. In the calculation of Alpha diversity index, all OTU within seven days of exposure were combined for each treatment group for statistical analysis. Chao1 index uses Chao1 algorithm to estimate the number of OTU in samples. In Fig. 4A, the Chao 1 index of the PET treatment group was significantly lower than that of the other four treatment groups ($P < 0.05$), indicating that the PET treatment group had fewer bacterial species compared with the other four treatment groups. Shannon and Simpson indices were used to reflect community diversity index. In Fig. 4B-C, Shannon index and Simpson index of PE treatment group were significantly higher than those of CK ($P < 0.05$), while Shannon index and Simpson index of PET treatment group were significantly lower than those of PE treatment group ($P < 0.05$). The results showed that compared with the control group, the richness and diversity of intestinal microbiota in PE treated group were higher. However, the alpha diversity of the PET treated mice was lower among the five treatment groups. Fig. 4I shows the amount of OTUs before exposure and on day 7 for the control group and all treatment groups, respectively. The amount of OTUs from day 1 to day 6 of exposure is shown in supporting information Fig. S5. Before microplastic exposure, the average amount of unique OTUs in intestinal microbiota of mice in each group was 217, and there was no significant difference among groups. The amount of CK was higher, and the amount of PE treatment group was lower. After 7 days of microplastic exposure, the amount of OTUs in the gut microbes of mice changed. The number of OTUs unique to PE and PP increased significantly by 31.2% and 15.3%, respectively, while the other treatment groups had the same obvious downward trend as the control group. These results indicated that the diversity of gut microbiota increased after exposure to PE and PP microplastics. This is the same as the conclusion in Fig. 4B-C that the Shannon index and Simpson index of PE treatment group are significantly higher than those of control group, and the richness and diversity

of intestinal flora are higher.

Non-metric multidimensional scaling analysis (NMDS) is a data analysis method that simplifies the research objects in multidimensional space to low-dimensional space for positioning, analysis and classification. We used the NMDS model to analyze the β diversity of the mouse gut microbiome, mainly because it can more accurately reflect the numerical ranking information of the distance matrix in the case of diverse species and a large number of species. In the analysis of all NMDS in this study, the stress < 0.2 , which has explanatory significance. Fig. S6 shows the NMDS analysis of CK and all treatment groups at the same time, which is difficult to analyze due to the large sample size. Therefore, we conducted non-metric multidimensional scaling analysis for CK and the five treatment groups respectively (Fig. 4D-H). There is no weight meaning in the horizontal and vertical axes of NMDS, and the difference between groups is seen according to the location distance between samples. In the analysis of CK and five treatment groups, the composition difference of intestinal microbiota between PS treatment group and control group was the largest (Fig. 4G), and the composition difference of intestinal microbiota between PVC treatment group and control group was the smallest (Fig. 4H).

3.5. The structure of gut microbiota after MPs exposure

Edible microplastic altered both the structure and composition of gut microbiota (Fig. 5). At the phylum level, more than 1% of the intestinal microbes were screened out (Fig. 5A), including eight bacteria, including Actinobacteria, Bacteroidetes, Deferribacteres, Epsilonbacteraeota, Firmicutes, Patescibacteria, Proteobacteria and Tenericutes respectively. Among them, Firmicutes and Bacteroidetes were dominant in phylum level. Then, ANOVA analysis was performed on bacteria with more than 1% phylum level to obtain bacteria in Fig. S7

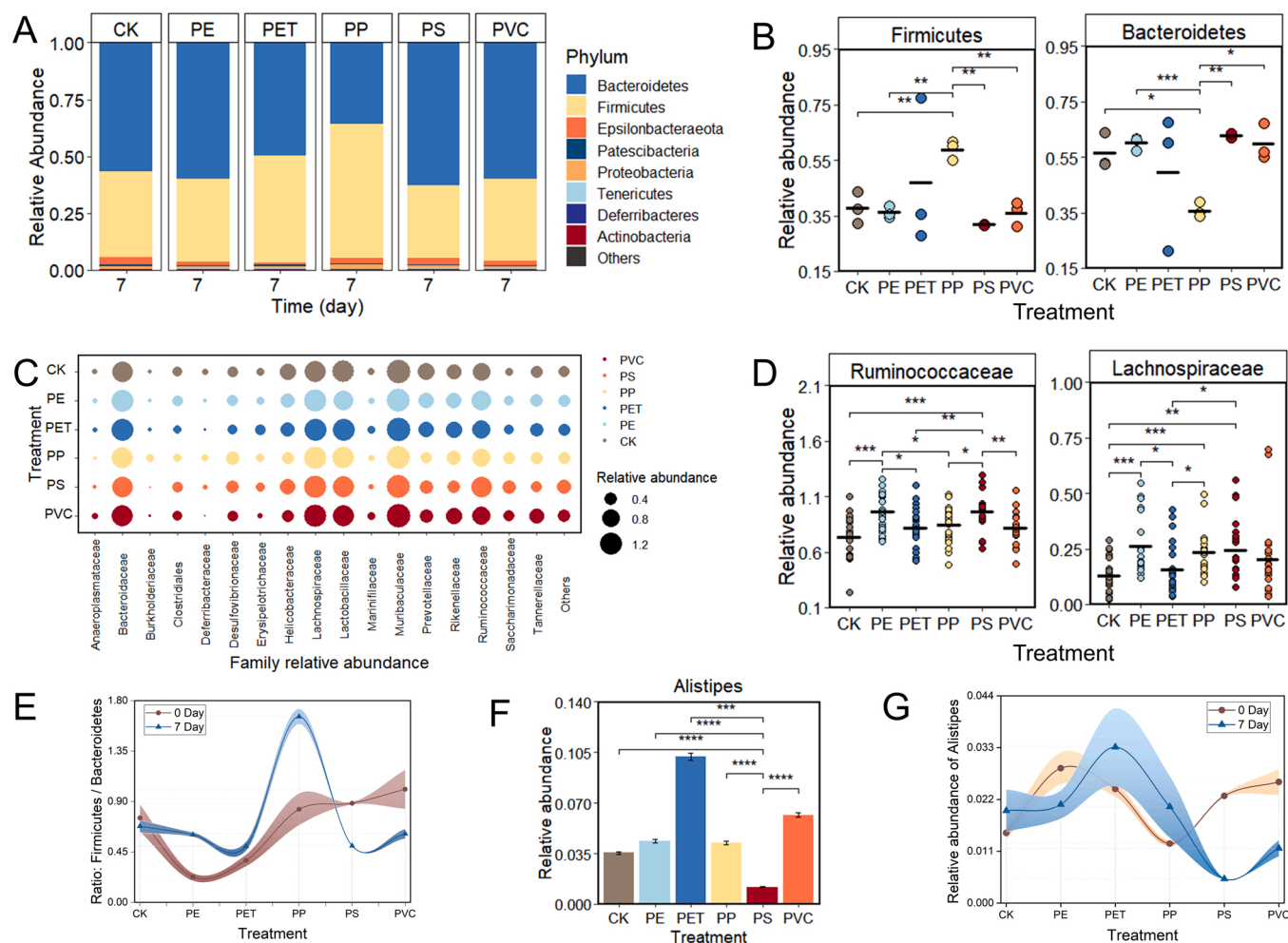


Fig. 5. Relative abundance of microflora at phylum level in mice exposed to microplastic (>1%) (A). At the phylum level, the abundance of Firmicutes and Bacteroidetes turned ($P < 0.05$) (B). Relative abundance of microflora at family level in mice exposed to microplastic (>1%) (C). At the family level, the abundance of Ruminococcaceae and Lachnospiraceae changed ($P < 0.05$) (D). The proportion of Firmicutes/Bacteroidetes in the intestinal flora of mice in all treatment groups before and after exposure (E). At the genus level, the abundance of *Alistipes* bacteria ($P < 0.05$) (F). Change trend of *Alistipes* bacteria in all treatment groups before and after exposure (G).

and Fig. 5B. According to Fig. 5B, it can be found that on the seventh day of exposure, compared with other treatment groups, Firmicutes and Bacteroidetes in PP treatment group were significantly increased and decreased, respectively. Fig. 5E shows the influence of edible microplastic on the ratio of Firmicutes/Bacteroidetes. The ratio of PE, PET and PP raised after exposure, while the ratio of PS and PVC treated groups decline.

Thereafter, further analysis of the species at the family level of more than 1% reveals more detailed structural changes of intestinal flora (Fig. 5C). A total of 17 species of bacteria are included, among which the bacteria with high relative abundance are Clostridiales_vadinBB60_group, Erysipelotrichaceae and Ruminococcaceae, all of which belong to Firmicheta. There are also Prevotellaceae and Rikenellaceae in Bacteroidetes and Desulfovibrionaceae in Proteobacteria. As shown in Fig. S8, these bacteria were remarkably different between the treatment groups. The bacterial abundance associated with inflammation at the family level is shown in Fig. 5D. Among the five kinds of microplastics, the abundance of Ruminococcaceae in PE and PS treatment groups increased significantly, and the relative abundance of Lachnospiraceae in PE, PP and PS also heightened significantly compared with the control group.

Finally, the most dominant 30 species of bacteria in the genus were heat map clustering (Fig. S9). According to Fig. S10 analysis, six species of bacteria were found to have significant differences between groups,

respectively are *Alistipes*, *Alloprevotella*, *Parabacteroides*, [Eubacterium] *Ventriosum_group*, *Enterorhabdus*, ASF356. It can be seen that *Alistipes* of inflammation-related bacteria were remarkably reduced compared with other treatment groups (Fig. 5F). In Fig. 5G, we compared the changes in the relative abundance of *Alistipes* bacteria in each treatment group before and after exposure. It can be found that after 7 days of exposure, *Alistipes* in PE and PVC groups reduced, especially in the PS treatment group, the abundance of *Alistipes* decreased the most.

4. Discussion

Presently, some researchers believe that plastic pollution that accumulates in the environment is "irreversible" (MacLeod et al., 2021). Nonetheless, in view of their low price and high practicability, plastic products have been widely used in daily life. At present, microplastics are ubiquitous in our natural environment. One study found that have shown that the abundance of microplastics in one Bay Area in the United States can reach about 700,000 particles per square kilometer. Microplastics have been found in some animals, indicating that the presence of microplastics has posed a threat to animal and human health. In the present study, we demonstrated that short-term exposure to PE, PET, PP, PS and PVC microplastics did not produce significant alterations in physiological and biochemical parameters of mice. However,

histopathology showed that damage to colonic tissue and exposure to microplastics significantly altered the abundance and structure of gut microbes in mice. In addition, there is no determination of the residual microplastics in mice, which is a deficiency of this paper. By observing the accumulation of microplastics in mice and the composition of intestinal microorganisms within 7 days, a joint analysis can be conducted to better understand the relationship between the effect of microplastics on intestinal microorganisms and the residual microplastics in the intestinal tract. In previous studies, we found that microplastics could be found in the organs of mice under both acute and subacute exposure (Meng et al., 2022; Xu et al., 2021). Therefore, we hypothesized that some microplastics remained in the gut of mice during our 7-day exposure experiment, and the altered intestinal microenvironment may have a certain impact on the composition and structure of intestinal microbes in mice. However, whether the residual microplastics in the gut are really related to the composition of intestinal microbes remains to be verified.

Body weight is a commonly used sensitive indicator in toxicology (Piao et al., 2013). After 7 days of exposure, the body weight of mice in all treatment groups and control group showed an upward trend, but there was no significant difference. Energy supply and storage are the main functions of TG in organisms, and they play an important role in the fixation and protection of viscera. TG is an important factor affecting cardiovascular diseases (Singh et al., 2009). Moreover, the TG level is closely related to liver function, and its imbalance can lead to the disorder of liver lipid metabolism (Yuan et al., 2020). Our study showed that the content of TG in mice treated with PVC and PET microplastics remarkably increased compared with the control group. In addition, no significant changes were found in TC, HDL-C and LDL-C levels.

Microplastic can cause damage to organic tissues such as lungs and enteric canal, mainly because they can pass through blood circulation and get into tissues (Guo et al., 2022; Jin et al., 2021b,a; Lu et al., 2021). Now there are studies that show that MPs could accumulate in the liver of a type of anchovies and cause tissue damage (Collard et al., 2017). In addition, other studies have reported that microplastic can damage tissues such as the bladder, kidneys and liver in mice (Wang et al., 2022; Xu et al., 2021). In this study, the H&E staining results showed pathological damages in colon tissues of mice in all groups except the control group. Microplastic cause structural conversions in the crypts of colon tissue, resulting in irregular surfaces, distorted glands and atrophy. In addition, the mice in the treatment group showed inflammatory pathological features, such as goblet cell reduction and inflammatory cell infiltration. These results suggest us that the colonic microenvironment of mice is disrupted in short-term microplastic exposure and accompanied by inflammation. Among them, PS treated mice had the highest proportion of inflammatory cells in all treatment groups.

Many studies have shown that promoting inflammation and oxidative stress are mutually reinforcing processes (Han et al., 2007). We further analyzed the response of mice to oxidative stress levels. In this study, the enzyme activities of GSH, SOD, POD and CAT of mice were affected to varying degrees after 7 days short-term exposure. As shown in Fig. 2, microplastic except PE enhanced the enzyme activity of GSH in all treatment groups, which might be a response to the excess reactive oxygen species stimulated by them in cells. MDA activity in all treatment groups raised compared with control group, indicating that lipid peroxidation occurred in colonic membrane. These cells then had elevated levels of reactive oxygen species that disrupt cellular homeostasis (Elsaesser and Howard, 2012). The results of our study are basically consistent with the previous findings that PS microplastic activate oxidative stress levels in the kidney and liver of mice (Li et al., 2021; Meng et al., 2022). In general, all the short-term exposure of each microplastic (PE, PET, PP, PS and PVC) can induce an imbalance of antioxidant defense system in mice.

Intestinal flora was existed as biological barriers and play a nutritional support, immune regulation and anti-tumor role for the organism (Gill et al., 2006). Studies have shown that changes in the structure of

the intestinal flora can cause metabolic disorders in the host, and even lead to disease (Thaiss et al., 2016). In the short-term exposure experiment, we studied the effects of microplastic on the intestinal microflora of mice. The diversity of microbiota in the gut of mice exposed to PET treatment reduce (Chao 1 index and Shannon-Wiener Index), but the Alpha diversity of microbiota in the gut of mice exposed to PE treatment heighten compared to controls (Shannon-Wiener index and Simpson Index). These results indicate that short-term exposure to microplastics (PE, PET, PP, PS and PVC) turned the microbial diversity of the intestinal flora to some degree. In addition, the intestinal microbiome structure of mice also changed after exposure. As can be seen from Fig. 4G, compared with other treatment groups, there are significant differences in sample analysis between the mice exposed to PS microplastic and the control mice. There are also researchers who are consistent with our research results. For example, Jin et al. (2019) also found that the intestinal microbiota of PS microplastic treated group is different from that of the control group when studying the effects of polystyrene microplastic on the intestinal barrier and microbiota of mice. Changes in some fixed bacterial can predict the health of organisms to a certain extent, especially associate with some disease. It has been shown that host intestinal flora exposure to MPs has an important impact on systemic disease (Feng et al., 2021; Yan et al., 2020). In terms of the composition of intestinal flora, it turned at all levels in the short-term exposure. At the family level, we found significant vary in the bacteria associated with inflammation compared with the control group (Fig. 5D). In both Ruminococcaceae and Lachnospiraceae, the abundance of PS treatment group increased significantly. Studies have shown that the abundance of Ruminococcaceae and Lachnospiraceae flora is associated with inflammatory enteritis (Clemente et al., 2018). The enhancement of anti-inflammatory bacteria indicates the inflamed status of host gut. In addition, at the genus level, the relative abundance of Alistipes in PS group declines remarkably. Researchers have found that the relative abundance of Alistipes in mice is reduced with colitis, so it is considered as a bacterium that can protect the host against colitis (Parker et al., 2020). In this study, the abundance of Alistipes in PS treated mice decreased significantly compared with all treatment groups, indicating a relatively high degree of inflammation. The abundance of inflammation-related bacteria at both family and genus levels showed significant vary in the PS treatment group compared with the other treatment groups. Through the quantitative analysis of pathological damage of mouse colon tissue in Fig. 2B, we found that the PS treatment group did have the most severe damage. Therefore, we believe that conversions in the composition and structure of intestinal microbes can reflect the intestinal injury of mice. Firmicutes and Bacteroidetes are present in over 90% of health organisms, and the Firmicutes/Bacteroidetes ratio is closely correlated with obesity. Previous studies have shown that the relative abundance of Bacteroidetes is associated with the body weight (Turnbaugh et al., 2006). A raise proportion of Firmicutes in the intestinal flora promotes nutrient uptake by the host and is associated to some extent with obesity (Ley et al., 2005). Bacteroidetes play an important role in maintaining intestinal physiological function, carbohydrate fermentation and human health (J. Xu et al., 2003). In this work, the body weight of mice in each treatment group was assessed and the serum indexes of mice was tested on a daily basis. The weight of mice in all treatment groups showed a slight trend of increase but no significant difference. Comparatively speaking, the slope of PET and PP groups was a little larger (Fig. 1B). In serum triglyceride index, PET and PVC groups had an obvious upward trend (Fig. 1D). The effect of microplastic on cholesterol in mice did not show up in the other three measures. Followed, the intestinal flora analysis showed the abundance of Firmicutes and Bacteroidetes, the dominant bacteria in intestinal microbes, turned at the phylum level under exposure to five kinds of microplastics. In particular, Firmicutes were remarkably increased and Bacteroidetes significantly decreased in the PP treatment group. As shown in Fig. 5E, the ratio of Firmicutes/Bacteroidetes in PP treatment group heighten rapidly, and the abundance of Firmicutes and

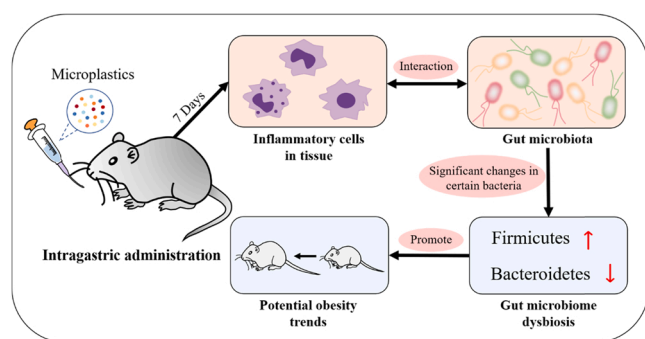


Fig. 6. Causal diagram of effects of microplastic exposure on mice.

Bacteroides in the PET treatment group were only lower than those in the PP treatment group. This is consistent with our above analysis results on physiological and biochemical indexes of mice, which may indicate the potential obesity problem of mice exposed to microplastic in a short period. Fig. 6 shows that exposure to microplastic increased infiltration of inflammatory cells, causing inflammation of colon tissue in mice. On the other hand, microplastic exposure significantly altered the structure of gut microbes, with some bacteria associated with inflammation showing significant vary. The emergence of inflammatory cells and the imbalance of intestinal microorganisms interact and reinforce each other. In addition, the significant changes of Firmicutes and Bacteroidetes in intestinal flora could reflect the potential obesity trends that could not be manifested under short-term exposure of microplastic.

5. Conclusion

In summary, the biological effects of acute exposure to PE, PET, PP, PS and PVC microplastics were investigated in intestinal tract of mice model. We found that the structural changes of intestinal microorganisms can reflect the subacute toxicity of MPs in short-term exposure. The toxic response of mice exposed to microplastic for a short period of time was mainly manifested in oxidative damage and histopathological damage as well as potential obesity problems. Our research fills in the knowledge gap of the toxicity comparison of various microplastics during short-term exposure. These findings provide new insights into the potential risks of multicomponent microplastics in acute exposure to mammals and may further provide new insights into the possible effects of multicomponent microplastics on human health.

CRediT authorship contribution statement

YL and TC conceived the idea and led the study design. LX, TL, YH and QT carried out these experiments. LX, XZ and ZL involved in the data analysis and wrote the manuscript. YL, HT and WY gave critical suggestions and contributed to the editing of the manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The authors are unable or have chosen not to specify which data has been used.

Acknowledgements

This study was supported by the National Natural Science Foundation of China (grant number 21825403, 21976194), the Scientific

Research Fund of Hunan Provincial Education Department (grant number 18B166, 18B171), the Key Research and Development Project of Hunan Province (grant number 2020NK2022), the Training Program for Excellent Young Innovators of Changsha (grant number kq1905065), the Youth Scientific Research Foundation of Central South University of Forestry and Technology (CSUFT) (grant number QJ2017001 A).

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ecoenv.2022.114194](https://doi.org/10.1016/j.ecoenv.2022.114194).

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